

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I **K.MAHESH BABU(192525122)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **K.MAHESH BABU(192525122)**

Assessment Tasks

Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful	Some relevant examples	Weak examples	No useful examples

		examples			
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Define Big Data and explain why traditional data processing approaches are often insufficient for Big Data environments.

Answer:

Big Data refers to extremely large, complex, and rapidly generated datasets that cannot be efficiently stored, processed, and analyzed using traditional data-processing systems.

Traditional systems are often insufficient because:

1. **Huge Volume:** Big Data may contain terabytes or petabytes of information.
2. **High Velocity:** Data can be generated continuously and at very high speed.
3. **Variety:** Data may be structured, semi-structured, or unstructured.
4. **Scalability:** A single computer may not have enough storage or processing power.
5. **Processing Time:** Processing very large datasets on one system can take a long time.
6. **Fault Tolerance:** Failure of one system can affect data availability.

Therefore, Big Data platforms use **distributed storage and parallel processing**, where data is divided among multiple computers and processed simultaneously.

Example: Social media platforms generate huge amounts of posts, images, videos, comments, and user activity every second. Traditional single-server systems may struggle to process such data efficiently.

2. Explain the 5Vs of Big Data with one practical example for any two Vs.

Answer:

The 5Vs of Big Data are:

V	Meaning	Explanation
Volume	Amount of data	Refers to the huge quantity of data generated and stored.
Velocity	Speed	Refers to how quickly data is generated, transmitted, and processed.
Variety	Different types	Refers to different formats such as text, images, video, JSON, etc.
Veracity	Accuracy	Refers to the reliability and quality of data.
Value	Usefulness	Refers to the useful information obtained from data analysis.

Practical Example – Velocity:

In online banking, thousands of transactions may occur every second. The system must process transaction information quickly to detect fraud and update account balances.

Practical Example – Variety:

A hospital may have patient records, medical images, laboratory reports, doctor's notes, and sensor readings. These represent different types and formats of data.

3. Differentiate structured, semi-structured, and unstructured data with examples.

Answer:

Type	Description	Examples
Structured Data	Data organized using a fixed schema, usually rows and columns.	Student database, bank transactions, employee records
Semi-Structured Data	Data without a strict table structure but containing tags, keys, or metadata.	JSON, XML, emails
Unstructured Data	Data without a predefined structure or schema.	Images, videos, audio, social-media posts

Structured Data:

It is easy to store and query using relational databases.

Semi-Structured Data:

It has some organizational structure but can have different fields between records.

Unstructured Data:

It requires specialized technologies and processing techniques because there is no fixed schema.

Example: In an online shopping system, customer details stored in a database are structured data, a JSON order document is semi-structured data, and product images are unstructured data.

4. Explain the role of distributed file systems in Big Data processing.

Answer:

A **Distributed File System (DFS)** stores large amounts of data across multiple computers or storage nodes instead of keeping everything on one machine.

Its important roles include:

1. **Scalability:** Additional machines can be added when more storage is required.
2. **Fault Tolerance:** Multiple copies of data can be maintained so that data remains available even if one machine fails.
3. **Parallel Processing:** Different parts of a dataset can be processed simultaneously.
4. **Large Storage Capacity:** Very large datasets can be stored across multiple machines.
5. **Data Availability:** Data can remain accessible even when individual nodes experience failures.

A distributed file system is therefore an important component of Big Data platforms because it provides **distributed storage, scalability, reliability, and support for parallel processing.**

Example: Hadoop Distributed File System (HDFS) distributes large files across multiple nodes in a Hadoop cluster.

5. List any three major security concerns in Big Data platforms and briefly explain each.

Answer:

1. Data Privacy

Big Data platforms may contain sensitive information such as customer details, financial information, and healthcare records. Unauthorized access can lead to privacy violations.

Solution: Use authentication, authorization, encryption, and access control.

2. Data Integrity

Data may be accidentally or intentionally modified, deleted, or corrupted. Incorrect data can produce incorrect analytical results.

Solution: Use data validation, integrity checks, backups, and audit logs.

3. Data Security

Data may be exposed while it is stored or transferred between systems.

Solution: Use encryption for data **at rest** and **in transit**, secure communication protocols, and proper access controls.

Other security concerns include insider threats, malware, weak passwords, insecure APIs, and poor system configuration.

6. What are data appliances and integration tools? Explain their role in Big Data workflows.

Answer:

Data appliances are specialized hardware or integrated systems designed for data storage, processing, and analytics. They combine computing, storage, networking, and software components to handle data-intensive workloads efficiently.

Integration tools are software tools that connect different data sources and help collect, transform, clean, and transfer data between systems.

A typical Big Data workflow is:

Data Sources → Data Integration/Ingestion → Distributed Storage → Data Processing → Analytics → Visualization/Decision Making

For example, a company may collect:

- Customer information from databases
- Website activity from application logs
- Sales information from business applications
- Sensor data from IoT devices

Integration tools bring these datasets together and prepare them for analysis.

Therefore, data appliances provide the **infrastructure**, while integration tools help **connect and prepare data** for Big Data processing.

7. Discuss two major challenges organizations face while implementing Big Data analytics.

Answer:

1. Data Management and Quality

Organizations collect data from many different sources and formats. The data may be incomplete, duplicated, inconsistent, or inaccurate.

Poor-quality data can produce unreliable analytical results.

Solution: Organizations need data cleaning, integration, validation, governance, and proper storage systems.

2. Security and Privacy

Big Data systems may contain sensitive business and customer information. Protecting this information across distributed systems is challenging.

Organizations must control who can access data and protect it from unauthorized users.

Solution: Use authentication, authorization, encryption, monitoring, auditing, and security policies.

Other challenges include:

- High implementation cost
- Lack of skilled professionals
- Complex infrastructure
- Scalability
- Choosing appropriate Big Data technologies

8. Write short notes on any one domain application of Big Data analytics.

Answer: Healthcare Application

Big Data analytics has many applications in the **healthcare domain**. Healthcare organizations generate large amounts of data from electronic health records, medical images, laboratory reports, prescriptions, wearable devices, and patient-monitoring systems.

Big Data analytics can be used for:

1. **Disease Prediction:** Identifying patterns and risk factors associated with diseases.
2. **Patient Monitoring:** Analyzing real-time information from medical devices and wearable sensors.
3. **Medical Image Analysis:** Supporting the analysis of CT, MRI, and other medical images.
4. **Hospital Management:** Predicting patient demand and planning hospital resources.
5. **Personalized Treatment:** Using patient information to support individualized treatment decisions.
6. **Risk Analysis:** Identifying patients who may require additional monitoring.

Example: A hospital can analyze historical patient records and monitoring data to identify patients who may be at higher risk of developing complications.

Big Data analytics can improve healthcare decision making, but healthcare data must be protected using strong **privacy and security measures**.

9. How do business drivers influence the adoption of Big Data solutions?

Answer:

Business drivers are the organizational needs and objectives that encourage companies to adopt Big Data technologies.

Important business drivers include:

1. Better Decision Making

Organizations can analyze large datasets to identify trends and make evidence-based decisions.

2. Customer Understanding

Customer behavior, preferences, feedback, and purchase history can be analyzed to provide better services.

3. Cost Reduction

Analytics can identify inefficient processes and help organizations reduce unnecessary expenses.

4. Risk Management

Organizations can analyze data to identify unusual patterns and potential risks.

5. Competitive Advantage

Companies can use data insights to respond quickly to market changes and customer requirements.

6. Revenue Growth

Data analysis can help identify new products, services, customers, and business opportunities.

Therefore, organizations adopt Big Data solutions when the expected **business value and benefits** justify the required investment in technology, infrastructure, and skilled personnel.

10. Explain the importance of Big Data analytics in decision making.

Answer:

Big Data analytics converts large and complex datasets into useful information and insights. It helps organizations make decisions based on **data and evidence** rather than assumptions alone.

Big Data analytics supports:

1. **Strategic Decisions:** Long-term decisions such as entering new markets or developing new products.
2. **Tactical Decisions:** Decisions related to resource allocation, marketing, and planning.
3. **Operational Decisions:** Daily decisions such as inventory management and service optimization.
4. **Predictive Analysis:** Using historical data to predict possible future events.
5. **Risk Detection:** Identifying unusual patterns and potential problems.
6. **Customer Analysis:** Understanding customer needs and preferences.
7. **Cost Optimization:** Finding inefficient processes and reducing unnecessary expenses.

Example: A retail company can analyze previous sales, customer behavior, and seasonal trends to decide how much inventory should be maintained.

Thus, Big Data analytics enables organizations to make **faster, more accurate, data-driven decisions** and identify new opportunities.